NEO Mathematics Lesson Guidance

Straight-Line Graphs - $y = mx + c$

Driving Question: Can a line tell the story of change?

Lesson at a Glance

Mathematical focus: The central idea is Start + Change. In $y = mx + c$, c tells us the start and m tells us the change. Equation, table, graph and context are different representations of the same relationship.

Stepping Stones: connect equations, tables and graphs; interpret gradient as rate of change; interpret the y-intercept as a starting value; recognise positive, negative and zero gradients; identify m and c; write equations of straight lines; interpret linear models in context.

Primary NEO Cornerstones: Movement, Connection and Reflection.

Suggested duration: 60-90 minutes, or two shorter sessions. Timings are invitations, not deadlines. Follow the learner's regulation, engagement and mathematical readiness.

The lesson does not need to be completed in one sitting. A learner may speak, type, draw, use the Scratchpad, work on paper or pause and return.

Supporting the mathematics

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Resources and materials

Essential

• A device capable of running the interactive lesson

• Paper or digital working space

Helpful

• Ruler

• Graph paper

• Pencil

• Calculator

• Mini-whiteboard

• Two colours for marking 'start' and 'change'

Optional extension or context

• String or wool to represent a line

• Masking tape for a physical axis

• Coordinate cards

• Sticky notes labelled m and c

• Printed blank coordinate grids

Suggested lesson journey

Connect - about 5 minutes

Ask the Driving Question. Invite possible meanings for rising, falling and horizontal lines. A learner's graph story can be the entry point before formal notation.

Gradient and Intercept Explorer - 15-20 minutes

Begin with $y = 2x + 3$. Change only m, return to $m = 2$, then change only c. Invite a coordinate-grid Scratchpad prediction before revealing the updated graph.

Representation Matcher - 10-15 minutes

Use the routine: find the start, find the change, then match. Encourage the learner to say or type 'The story starts at ___ and changes by ___'.

Pause - flexible

Graphs can contain significant visual information. Close other panels, zoom, use paper, focus on one equation feature or return later.

Line Detective - 10-15 minutes

Use change in y divided by change in x before requiring formal gradient notation. Invite a gradient triangle on the Scratchpad.

Human Graph Story Builder - 15-20 minutes

Ask what happens before x changes, what changes for each increase of 1, whether the model can continue forever, and where it might stop being realistic.

Practise and Reflect - flexible

Use the progressive Practice Companion. The final portfolio piece can draw on the learner's story model.

Common misconceptions and supportive responses

Mixing up m and c

The learner identifies 3 as the intercept in $y = 3x + 2$.

Possible prompt: Set x equal to zero. What is y? Which number describes the start?

Negative gradient means negative y-values

The learner confuses a negative rate with a negative location.

Possible prompt: Does negative gradient describe where the line is, or how it is changing?

Steeper means higher

The learner compares vertical position rather than rate of change.

Possible prompt: Are we comparing starting point or rate of change?

Treating the graph as a picture

The learner focuses on overall shape without coordinates.

Possible prompt: What happens to y when x increases by exactly 1?

Assuming the model continues forever

The learner treats a mathematical line as a literal real-world future.

Possible prompt: Where would this line stop being a sensible model of the situation?

Cornerstone guidance

Connection

Mathematical role: Equation, table, graph and story are connected representations.

Notice or invite: Notice when the learner translates from one representation to another.

Movement

Mathematical role: Gradient describes mathematical change.

Notice or invite: Use the dynamic graph and prediction Scratchpad to make movement visible.

Creativity

Mathematical role: The learner constructs a meaningful linear model.

Notice or invite: Invite contexts that matter to the learner.

Reflection

Mathematical role: The learner explains m and c and examines model limitations.

Notice or invite: Ask what the model shows and what it hides.

Rest

Mathematical role: The graph can be read as Start. Change. Check.

Notice or invite: Reduce visual load and work with one feature at a time.

Nutrition

Mathematical role: Planning examples can connect linear models to food, event and resource decisions.

Notice or invite: Use Nutrition contextually; do not force it as the structural maths lens.

Regulation, access and re-engagement

If the learner does not want to speak: allow typed responses, pointing, changing interactive values, drawing or private Scratchpad work. Watching and testing values can still be mathematical engagement.

If the learner is worried about being wrong: invite a prediction rather than an answer. Use language such as, 'Let's test that idea' or 'What did the graph/recipe show us?'

If the learner becomes overwhelmed: reduce the visual field, close other panels, use the Rest routine, work with one quantity or one graph feature, or pause the session.

If there are gaps in prior knowledge: use the interactive as a concrete starting point. Do not require fluency before exploration. Record the gap as useful curriculum information rather than failure.

If the learner finishes quickly: invite variation, comparison, proof, model limitations or a learner-generated example rather than simply more of the same.

Evidence and mastery

Emerging - Identifies parts of an equation or reads points with support.

Developing - Identifies gradient and intercept in familiar forms and completes tables.

Secure - Connects m and c to graph features and interprets them in context.

Mastering - Creates, compares and evaluates linear models and communicates relationships across representations.

Possible evidence includes an explanation, interactive choices, a Scratchpad image, completed table, verbal teach-back, corrected misconception, model or final portfolio piece.

The NEO Maths Scratchpad

The embedded Scratchpad is a thinking space. It is not a compulsory evidence task. The learner can use blank paper, square grid, coordinate grid or a lesson-specific template. Undo, redo and eraser controls are intentionally available so that revising thinking feels normal.

Suggested facilitation language

> Notice first. Ask second. Explain only as much as the learner needs to keep thinking.

Useful prompts include: 'What are you noticing?', 'What stayed the same?', 'What changed?', 'Can you predict before we test?', 'What would convince you?', and 'Where might this model stop working?'